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AI for Bharat Hackathon

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Team Name : Team Magma

Team Leader Name : Saqib Ahmed K

Problem Statement : Build an AI-powered solution that helps people learn faster, work smarter, or become more productive while building or understanding technology.

Brief about the Idea:

CodeTutor AI is an AI-powered learning assistant designed specially for students, self-taught developers, and early career engineers in India who are learning to code or building tech projects, It acts like a 24/7 personal coding mentor which helping users understand confusing code, debug errors, simplify technical documentation and improve their development window all through a simple chat like interface.

Built on AWS cloud services and powered by generative AI (Amazon Bedrock), CodeTutor doesn't just give answers it teaches concepts. Whether you're stuck on a NullPointerException, trying to understand a React hook or overwhelmed by API docs, CodeTutor breaks it down in a clear, beginner-friendly language so you learn faster and build smarter.

It's not just another code generator, it is a learning accelerator for the next generation of Indian developers.

Your solution should be able to explain the following:

- **How different is it from any of the other existing ideas?**
  - Most AI coding tools (e.g., GitHub Copilot) generate code but don't teach understanding.
  - Along with understanding our code also recommends suitable remembering techniques and list of real-life applications of any given topics.
  - Search engines and forums require users to phrase perfect queries a major barrier for beginners.
  - CodeTutor works context-first: paste code → get explanation, paste error → get fix. No prompt engineering needed.
  - Designed specifically for learners in India, with emphasis on clarity over cleverness, and education over automation.

Your solution should be able to explain the following:

- How will it be able to solve the problem?

1. **Accelerates learning:** Turns hours of Googling into seconds of guided insight.
2. **Reduces frustration:** Instantly decodes cryptic error messages and complex syntax.
3. **Democratizes mentorship:** Brings expert-level guidance to students in Tier 2/3 cities with limited access to coaching.
4. **Boosts productivity:** Developers stay in flow—no context switching between IDE, browser, and notes.

Your solution should be able to explain the following:

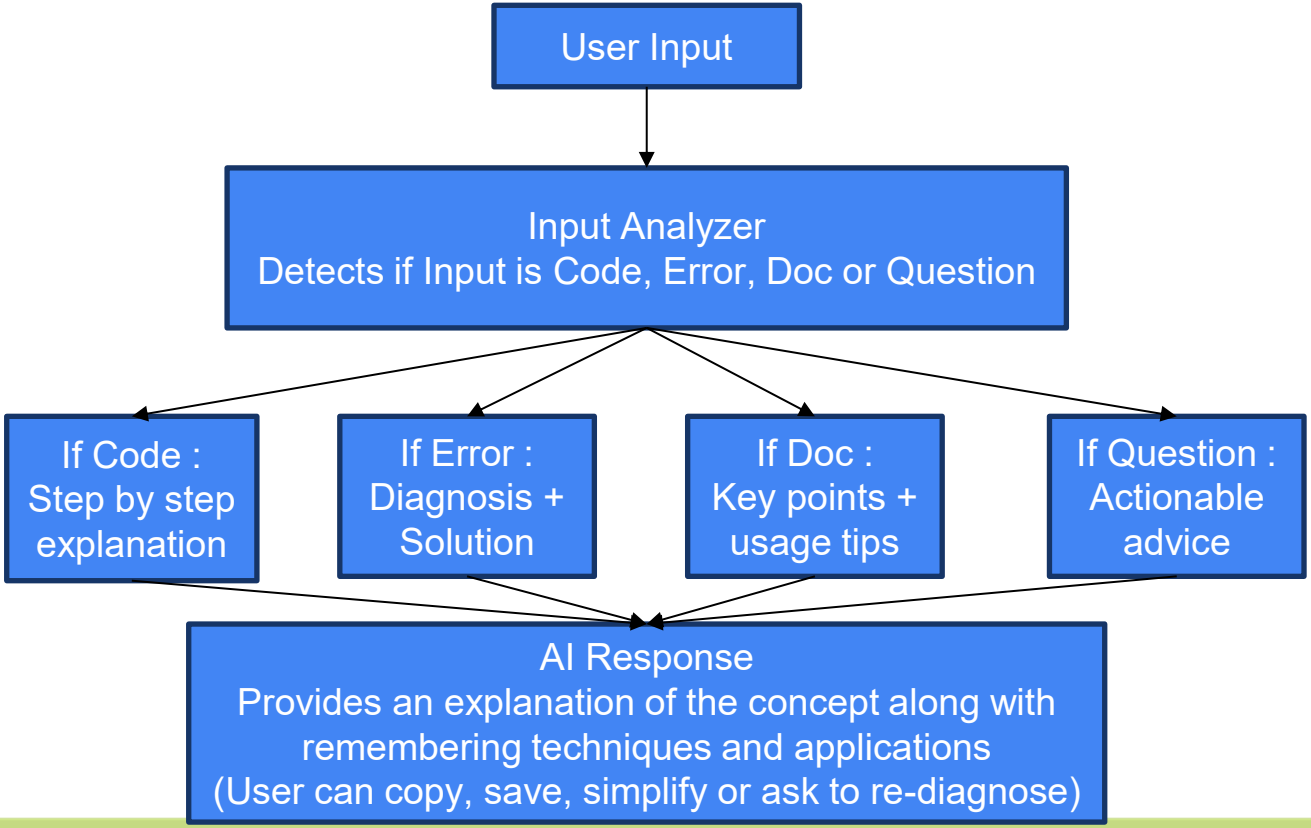
- USP of the proposed solution

"**From confusion to clarity in one click**."
CodeTutor AI is the only tool that combines code explanation, debugging, documentation summarization, and workflow coaching into a single, beginner-optimized experience—built on secure, scalable AWS infrastructure for India's next million developers.

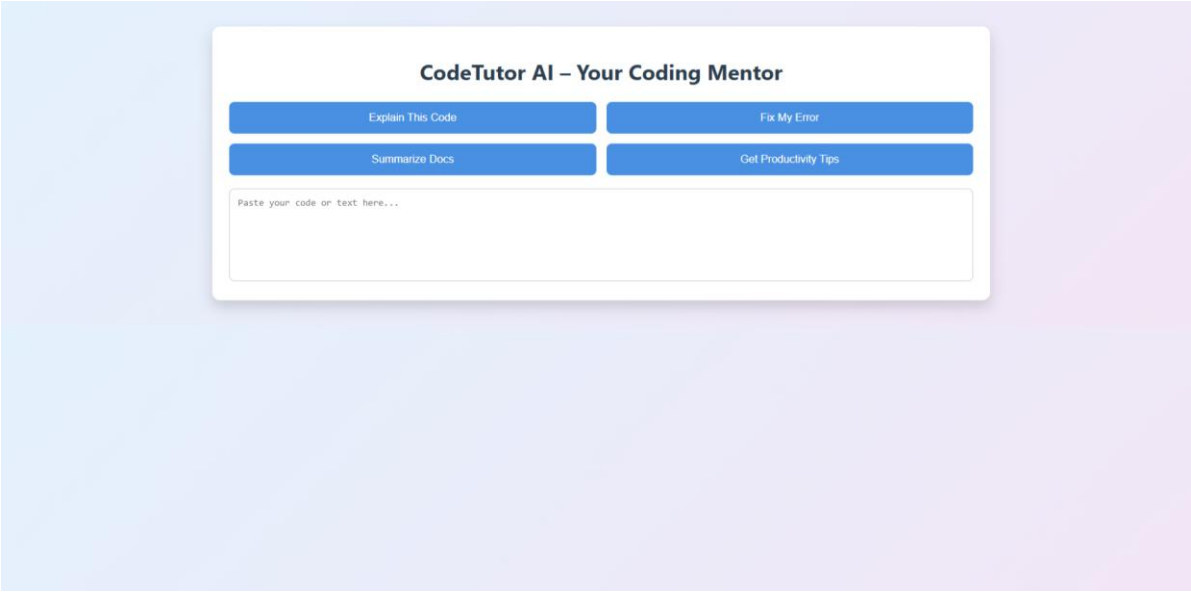
List of features offered by the solution

1. **Code Explainer :** Paste any code snippet → get a line-by-line breakdown in simple English.
2. **Bug Decoder :** Paste an error message (with or without code) → get root cause + fix.
3. **Doc Summarizer :** Paste technical documentation → get a concise, structured TL;DR.
4. **Workflow Coach :** Ask “How do I improve?” → get personalized productivity tips.
5. **Beginner Mode :** Simplifies responses for new learners (toggle on/off).
6. **Active Learning Techniques :** A combination of understanding, learning and execution.
7. **Future Implementation :** Regional language support via Amazon Translate (Hindi, Tamil, Bengali, etc.).

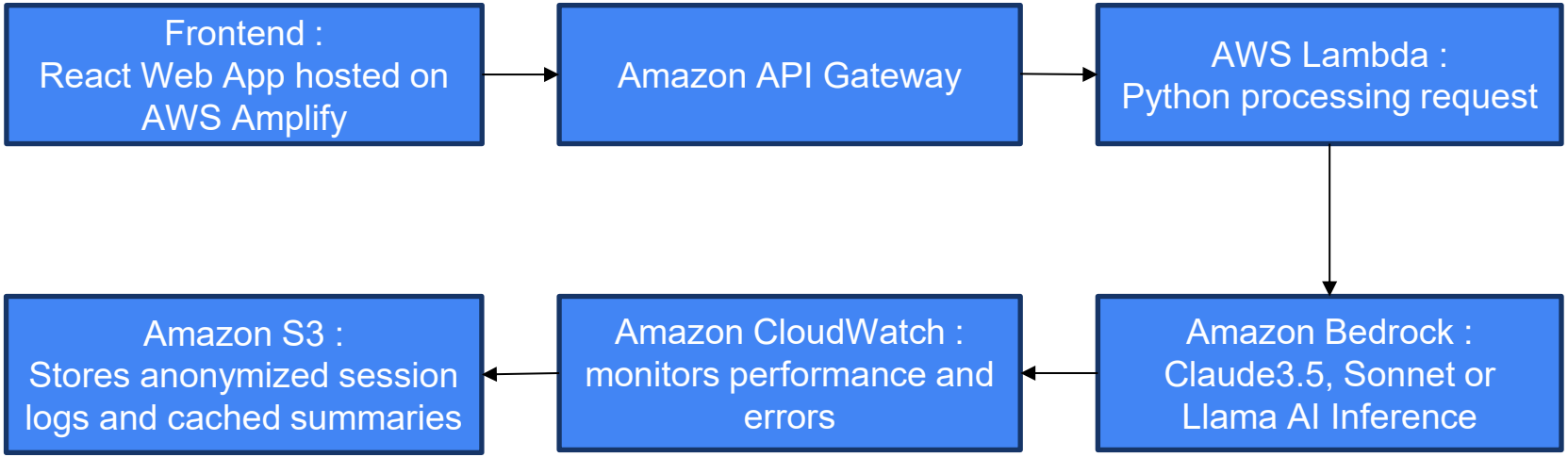
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:



Technologies to be used in the solution:

1. **Frontend**: React.js + CSS
2. **Hosting**: AWS Amplify
3. **Backend**: AWS Lambda (serverless functions)
4. **AI Engine**: Amazon Bedrock (Claude 3.5 or Llama 3)
5. **API Management**: Amazon API Gateway
6. **Storage**: Amazon S3
7. **Monitoring**: Amazon CloudWatch
8. **Future Implementation:** Amazon Translate for regional language support

Add as per the requirements for the hackathon:

- CodeTutor AI directly advances Digital India, Skill India, and Startup India by:
- Making high-quality tech education accessible, affordable, and inclusive
- Empowering students in non-metro regions to compete globally
- Reducing reliance on expensive coaching or urban tech hubs
- Building a self-reliant developer ecosystem across Bharat

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Thank You